

# Kumar Gaurav

Software Engineer | AI/ML Systems & Backend

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## Skills

**Languages:** Python, TypeScript, JavaScript, Go, SQL, Bash, C++ (competitive programming)

**ML & Data:** scikit-learn, LightGBM, pandas, NumPy, SciPy, feature engineering, cross-validation, NLP (TF-IDF)

**LLM Systems:** multi-agent orchestration, structured outputs (Zod), prompt-injection defense, Gemini API, Temporal

**Backend:** FastAPI, Node.js, Express.js, Next.js, React, Socket.IO, WebSockets, gRPC, Kafka, REST APIs

**Data & DevOps:** PostgreSQL, Redis, Prisma, Docker, GitHub Actions (CI/CD), Git, Linux

**Testing:** k6 (load testing), Vitest, Supertest, Go testing (race detector)

## Education

**Birla Institute of Technology, Mesra | Ranchi, Jharkhand**

Expected May 2027

B.Tech in Computer Science and Engineering, CGPA: 7.88/10

## Experience

**Backend Engineer | SoilSense (Soil Suitability Platform)**

Mar 2026 – Present

*Team project, Schneider Electric Hackathon 2026 | Python, FastAPI, pandas, Next.js, Gemini API | GitHub*

- Wrote the crop-suitability rules engine over a 2,200-sample, 22-crop dataset, with tolerance bands (10% on NPK, 0.3 on pH) that map each deficiency to an organic amendment.
- Optimized the all-crops check to load the dataset once instead of once per crop, making the 22-crop scan **2.3× faster** (48 ms to 21 ms) and replacing 22 client requests with one.
- Hardened the Gemini analysis route against prompt injection by allow-listing and type-casting the 5 expected sensor fields, so injected keys never reach the prompt.

**Open Source Contributor | Oppia, FOSSology**

Jan 2026 – Present

*3 merged upstream PRs | TypeScript, Angular, ESLint, CI/DevOps | Remote*

- Merged a custom ESLint rule into Oppia (6.8k stars) that fails CI on undocumented **ViewEncapsulation.None**, a recurring cause of CSS leaking across components; shipped it with 10 tests and a 41-file legacy allowlist to keep CI green.
- Earned merge access to review and land PRs for Oppia's Dev Workflow team after 2 PRs passed maintainer review.
- Migrated FOSSology's Vagrant dev environment off end-of-life Ubuntu 20.04 to 22.04 to match its CI runners.

**Research Intern | Birla Institute of Technology, Mesra**

May 2026 – Jul 2026

*NLP / Content Moderation | Python, scikit-learn, LightGBM, pandas | GitHub*

- Developed a 4-class comment-moderation classifier on 198k comments, combining word/char TF-IDF with 36 engineered signals (threat and identity lexicons, intent regexes, vote-controversy ratios) in an 80k-feature matrix.
- Trained Naive Bayes, L1 Logistic Regression and LightGBM with 5-fold stratified CV and class weights; the engineered features lifted a linear model from 0.54 to 0.82 macro-F1.
- Blended LightGBM and Logistic Regression with weights found by simulated annealing and per-class thresholds tuned with Nelder-Mead, reaching **0.836 macro-F1**, and raised F1 on the rare threat class (2.8% of data) from 0.23 to 0.65.

## Projects

**Loom | Multi-Agent LLM Orchestration Platform**

Aug 2026 – Sep 2026

*github.com/D3S-Gaurav/Loom | TypeScript, Next.js, Go, Kafka, gRPC, Redis, PostgreSQL, Temporal*

- Architected an LLM orchestrator whose planner emits a schema-validated task DAG; workers run independent tasks in parallel, and validator rejections trigger focused revisions (0–2 per run by mode).
- Capped LLM context cost with per-mode budgets on every hand-off (worker output, validator feedback, a 12k–48k-character synthesis corpus), shared fairly so one long answer cannot crowd out the rest.
- Treated attachments as untrusted input: GitHub fetches accept only exact repository URLs (no SSRF), and images reach agents only as vision-model text told to treat visible instructions as data.
- Streamed run events through Kafka and a Go service to a live graph UI over gRPC and WebSocket, with sequence-numbered replay so reconnecting clients never apply an event twice.

**ChatFlow | Real-Time Group Chat**

Jun 2026 – Jul 2026

*github.com/D3S-Gaurav/chat-fullstack | TypeScript, Node.js, Express, Socket.IO, Prisma, PostgreSQL, k6*

- Load-tested with k6 to **5,000 concurrent sockets** and **~97k message deliveries/s with zero loss** on one Node process (p95 40 ms at 1,000 sockets); traced the ceiling to the single-threaded broadcast.
- Designed the backend around a typed Socket.IO event contract with Zod validation on every payload, JWT socket auth and role-based access; 32 integration tests run in CI against PostgreSQL 17.

## Achievements

- Ranked 682 of 17,000+ participants (top 4%) in Codeforces Round 1102 (Div. 2); max rating 1306.
- Solved 600+ DSA problems across LeetCode, Codeforces, CodeChef and GeeksforGeeks.
- Finalist, SEBI Hackathon, a national-level hackathon hosted by the Securities and Exchange Board of India.
- Achieved Gold Level on WorldQuant BRAIN with 10,000+ points across 8 peer-reviewed alphas.